

CAISO Energy Demand Forecasting

*Statistical Forecasting, Weather Integration, Machine Learning,
and Interactive Decision Support*

Internship Final Project Report

Prepared from two analytical project phases:

Notebook 01 - Energy Demand Exploration & Statistical Forecasting

Notebook 02 - Weather Integration & Machine Learning

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Primary system: California Independent System Operator (CAISO)

Executive Summary

This project developed an end-to-end framework for forecasting hourly CAISO electricity demand using historical demand, weather observations, temporal structure, and machine-learning methods. The work was completed in two phases. **Phase 1** established a statistical forecasting baseline using SARIMA, Prophet, and a Hybrid SARIMA-Prophet approach. **Phase 2** integrated weather observations from five California locations, engineered temporal and lagged-demand features, and evaluated Random Forest and XGBoost models. **The main result is a clear progression in forecasting capability.** Among the Phase 1 statistical models, SARIMA produced the best overall accuracy with MAE = 1,214.92 MWh and MAPE = 5.25%, while the Hybrid model achieved the lowest RMSE = 1,545.61 MWh. In Phase 2, weather alone was weak (Random Forest $R^2 = 0.0186$), but adding time-based variables, lagged demand, rolling statistics, and weather features substantially improved performance. The Enhanced Random Forest achieved MAE = 398.17 MW, RMSE = 526.51 MW, MAPE = 1.67%, and $R^2 = 0.9475$. XGBoost was the strongest Phase 2 model, achieving MAE = 354.48 MW, RMSE = 461.88 MW, MAPE = 1.50%, and $R^2 = 0.9596$.

Feature-importance analysis showed that recent historical demand – particularly the one-hour and 24-hour lag features – was more influential than any single weather variable. This indicates that short-term electricity demand is driven primarily by recurring consumption behavior and recent system conditions, while weather and calendar variables provide complementary information. The project therefore demonstrates the value of combining time-series reasoning with feature-rich machine learning for operational demand forecasting.

1. Project Background and Objective

Reliable short-term electricity-demand forecasts help grid operators anticipate system load, schedule generation, manage reserves, and reduce the risk of supply-demand imbalance. California demand is strongly seasonal and can also be affected by weather conditions, especially summer heat and cooling requirements. The project was designed to move from a traditional demand-only forecasting baseline toward a broader forecasting framework that integrates external and engineered predictors.

1.1 Primary Objective

To develop and evaluate forecasting approaches for hourly CAISO electricity demand and identify a model that balances predictive accuracy, interpretability, and practical usefulness for a dashboard-based decision-support workflow.

1.2 Analytical Questions

- What recurring temporal patterns are present in CAISO hourly electricity demand?
- How well do SARIMA, Prophet, and a Hybrid model forecast demand using historical demand alone?
- How much predictive value does weather provide by itself?
- How much improvement is achieved by adding temporal features, demand lags, and rolling statistics?
- Which final model performs best in the weather-enhanced machine-learning phase?
- Which features drive the final model predictions?

2. Data and Preparation

2.1 Electricity-Demand Data

The demand dataset contains 8,760 hourly CAISO observations covering the full 2023 calendar year. The analysis retained the timestamp and demand variable, corrected the small number of missing demand observations, sorted the series chronologically, and verified timestamp integrity before modeling.

2.2 Weather Data

The second phase integrated hourly weather observations representing five California locations: San Francisco, Los Angeles, Sacramento, Fresno, and San Diego. Weather variables included temperature, humidity, precipitation, wind speed, and atmospheric pressure. Statewide temperature summary features were also constructed from the location-level observations.

2.3 Integrated Modeling Dataset

After weather integration and cleaning, the project retained a complete hourly dataset for 2023. Missing precipitation values were handled, remaining gaps were interpolated where appropriate, duplicate timestamps were checked, and the data were maintained in chronological order for time-aware modeling.

3. Project Workflow

Stage	Main Activities	Outcome
Exploration	Hourly demand cleaning; decomposition; stationarity tests	Established daily, weekly, and annual demand structure
Phase 1	SARIMA, Prophet, Hybrid forecasting	Statistical baseline and model comparison
Weather integration	Five California weather locations; merged hourly features	Expanded external predictor set
Feature engineering	Time features, demand lags, rolling averages	Captured temporal dependence and recent load behavior
Phase 2	Weather-only RF, Enhanced RF, XGBoost	Strong machine-learning forecasting performance
Delivery	Streamlit dashboard; natural-language commentary	Interactive communication of forecasts and model results

4. Phase 1 - Energy Demand Exploration and Statistical Forecasting

4.1 Exploratory Time-Series Findings

Seasonal decomposition showed strong daily (24-hour) and weekly (168-hour) cycles. Demand was generally lower during winter and spring, increased substantially during summer, and then returned toward baseline levels later in the year. The analysis also identified larger residual fluctuations during high-demand summer periods, supporting the hypothesis that external weather conditions contribute to unexplained demand variation.

4.2 Stationarity and SARIMA Development

ADF and KPSS testing was used to assess stationarity. Weekly seasonal differencing improved stationarity, while a 24-hour seasonal SARIMA specification was ultimately adopted as a computationally practical baseline for full-year hourly modeling. The evaluation model used a chronological split, training on January through October 2023 and testing on November through December 2023.

4.3 Phase 1 Model Results

Model	MAE (MWh)	RMSE (MWh)	MAPE (%)	Interpretation
SARIMA	1,214.92	1,604.52	5.25	Best overall average accuracy
Prophet	3,047.00	3,728.00	13.05	Captured broader trend/seasonality but weaker hourly accuracy
Hybrid SARIMA + Prophet	1,261.45	1,545.61	5.35	Lowest RMSE; reduced larger errors

Phase 1 conclusion: SARIMA was selected as the strongest traditional statistical model because it achieved the lowest MAE and MAPE. The Hybrid approach remained useful because its lower RMSE indicates improved robustness against larger forecast errors.

5. Phase 2 - Weather Integration and Machine Learning

5.1 Weather-Only Baseline

The initial Random Forest used weather variables to test whether meteorological information alone could explain hourly electricity demand. The model achieved MAE = 1,939.99 MW, RMSE = 2,403.55 MW, and $R^2 = 0.0186$. This low R^2 demonstrated that weather is relevant but insufficient by itself for accurate hourly load forecasting.

5.2 Feature Engineering

The enhanced feature set combined weather variables with calendar and historical-demand information. The core engineered predictors included hour of day, day of week, month, quarter, day of year, weekend indicator, demand lags at 1, 24, and 168 hours, and rolling demand statistics. This structure allowed the models to learn both external conditions and recurring load behavior.

- Temporal features: hour, day of week, month, quarter, day of year, weekend indicator.
- Historical demand features: 1-hour, 24-hour, and 168-hour lags.
- Rolling demand features: short- and weekly-window demand summaries.
- Weather features: temperature, humidity, wind, precipitation, pressure, and statewide temperature summaries.

5.3 Enhanced Random Forest Results

<i>Metric</i>	<i>Weather-only RF</i>	<i>Enhanced RF</i>	<i>Change</i>
MAE	1,939.99 MW	398.17 MW	79.5% reduction
RMSE	2,403.55 MW	526.51 MW	78.1% reduction
MAPE	N/A	1.67%	Strong forecasting accuracy
R ²	0.0186	0.9475	Large explanatory improvement

The improvement confirms that historical demand and temporal structure are essential predictors. Weather adds useful context, but the strongest gains came from combining it with recent demand behavior and calendar structure.

5.4 XGBoost Results

<i>Model</i>	<i>MAE (MW)</i>	<i>RMSE (MW)</i>	<i>MAPE (%)</i>	<i>R²</i>
Enhanced Random Forest	398.17	526.51	1.67	0.9475
XGBoost	354.48	461.88	1.50	0.9596

XGBoost outperformed the Enhanced Random Forest across all Phase 2 evaluation metrics. It therefore represents the strongest weather-enhanced machine-learning model in the project and the preferred final forecasting model within the Phase 2 framework.

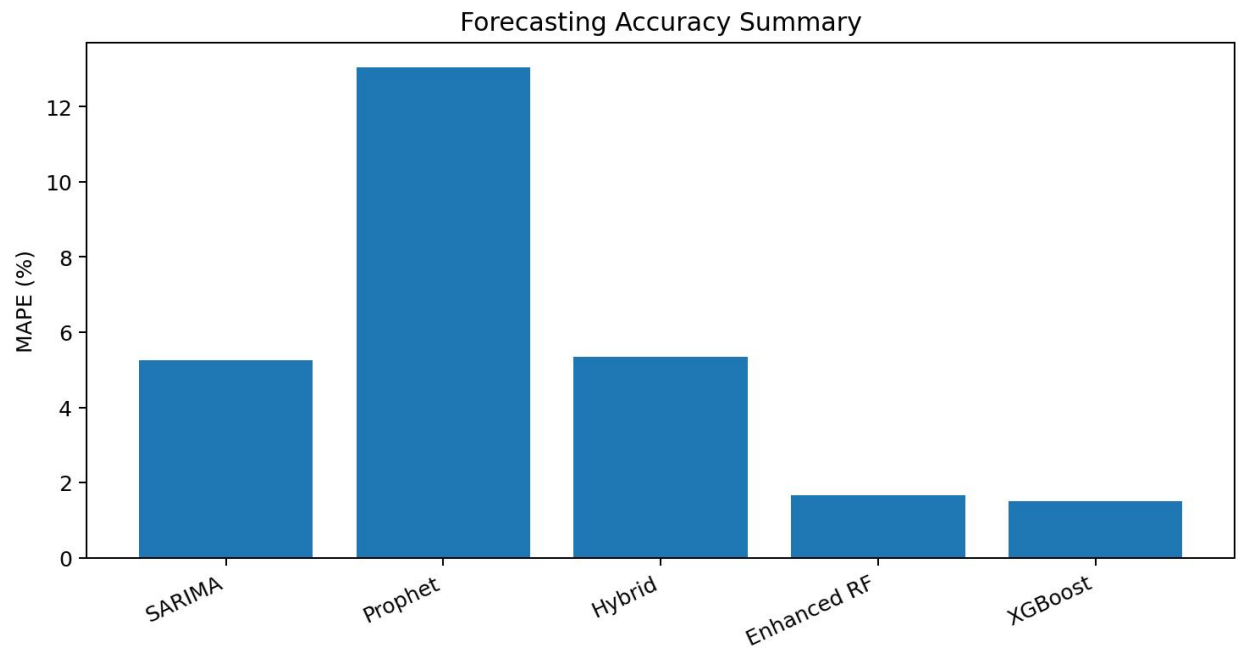


Figure 1. MAPE summary for models with reported MAPE values. Phase 1 and Phase 2 use different evaluation splits; cross-phase comparisons should therefore be interpreted as directional rather than strictly like-for-like.

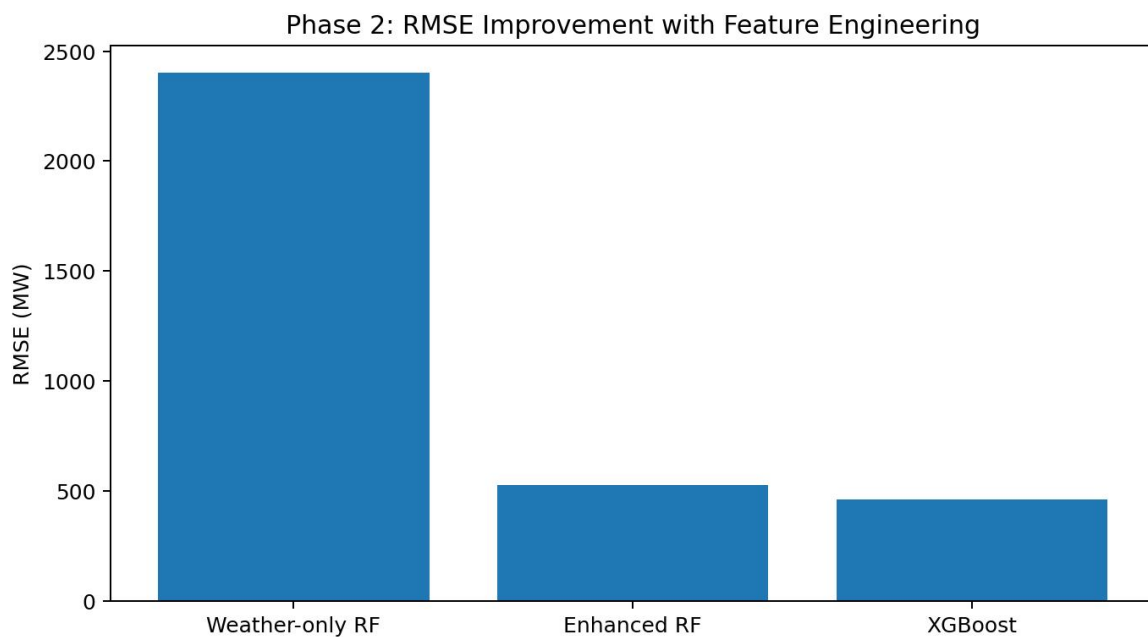


Figure 2. Phase 2 RMSE improvement from weather-only Random Forest to feature-engineered models.

6. Feature Importance and Interpretation

XGBoost feature importance showed that the strongest predictor was the one-hour demand lag, followed by the 24-hour lag. Weekend status, hour of day, day of week, and

the 168-hour lag also contributed. Weather variables such as temperature and precipitation improved the model, but their individual importance was smaller than the most recent historical-demand features.

Rank	Feature	Importance
1	Demand lag - 1 hour	0.8308
2	Demand lag - 24 hours	0.0755
3	Weekend indicator	0.0269
4	Hour of day	0.0177
5	Day of week	0.0080
6	Demand lag - 168 hours	0.0065
7	Quarter	0.0060
8	Maximum temperature	0.0023

Interpretation: electricity demand is highly autoregressive. Recent load conditions encode much of the information needed for the next-hour forecast. Calendar effects help the model distinguish recurring consumption patterns, while weather provides incremental information that is especially relevant during temperature-driven demand changes.

7. Overall Model Assessment

The two project phases answer different but complementary questions. Phase 1 establishes how far historical demand alone can go using classical time-series methods. Phase 2 tests whether a broader feature set can improve predictive accuracy through machine learning.

Model	Phase	MAE	RMSE	MAPE	R ²
SARIMA	1	1,214.92 MWh	1,604.52 MWh	5.25%	–
Prophet	1	3,047.00 MWh	3,728.00 MWh	13.05%	–

Model	Phase	MAE	RMSE	MAPE	R ²
Hybrid	1	1,261.45 MWh	1,545.61 MWh	5.35%	-
Weather-only Random Forest	2	1,939.99 MW	2,403.55 MW	-	0.0186
Enhanced Random Forest	2	398.17 MW	526.51 MW	1.67%	0.9475
XGBoost	2	354.48 MW	461.88 MW	1.50%	0.9596

Comparison note: Phase 1 and Phase 2 were evaluated with different chronological train/test windows, so the absolute metrics should not be treated as a perfectly controlled head-to-head experiment across phases. The project conclusion is based on the consistent Phase 2 comparison between the Enhanced Random Forest and XGBoost, together with the strong reduction in error after feature engineering.

8. Dashboard and Interpretation Layer

The project also includes a Streamlit dashboard to communicate model performance and forecasting outputs in an accessible format. The dashboard is intended to support model selection, metric review, historical/forecast visualization, and plain-English interpretation of forecasting results.

The current prototype uses pre-generated natural-language commentary rather than requiring a paid live LLM API. The architecture can be extended later to a live LLM service if desired. This design keeps the project reproducible while preserving the decision-support concept of translating model outputs into understandable operational commentary.

9. Limitations

- The analysis uses only one year (2023) of demand and weather data, which limits assessment across multiple climate and market regimes.
- Weather observations come from five representative California locations and may not fully capture localized conditions across the entire CAISO service region.

- Variables such as electricity prices, renewable generation, holidays, economic activity, population changes, and detailed grid operations were not included.
- Hyperparameter optimization was limited; systematic tuning and time-series cross-validation could improve robustness.
- Extreme events such as wildfires, prolonged heatwaves, outages, or emergency grid conditions were not explicitly modeled.
- Cross-phase metric comparison is not strictly like-for-like because the statistical and machine-learning phases used different chronological evaluation splits.

10. Future Work and Recommendations

1. Expand the dataset to multiple years of CAISO demand and weather observations.
2. Add exogenous variables such as renewable generation, electricity prices, holidays, economic indicators, and grid operational data.
3. Evaluate SARIMAX with exogenous regressors as a direct statistical comparison to the weather-enhanced machine-learning models.
4. Apply systematic hyperparameter optimization and time-series cross-validation.
5. Investigate sequence and deep-learning models such as LSTM, GRU, Transformer-based architectures, and Temporal Fusion Transformers.
6. Extend the dashboard toward live model serving, real-time weather feeds, and optional live LLM commentary.

11. Final Project Conclusion

This project demonstrates a complete progression from exploratory time-series analysis to statistical forecasting, weather integration, feature engineering, machine learning, and interactive result communication. The Phase 1 analysis established SARIMA as the strongest traditional statistical forecasting model for hourly CAISO demand, with MAPE = 5.25%. The Hybrid SARIMA-Prophet approach reduced larger errors and achieved the lowest Phase 1 RMSE.

Phase 2 showed that weather alone is not sufficient for accurate hourly demand forecasting. The major improvement came from combining weather with temporal structure and recent demand history. The Enhanced Random Forest produced strong performance, and XGBoost improved further to MAE = 354.48 MW, RMSE = 461.88 MW, MAPE = 1.50%, and $R^2 = 0.9596$.

The final analytical insight is that recent electricity-demand history is the dominant source of predictive information, while weather and calendar variables provide complementary value. Within the scope of the Phase 2 evaluation, XGBoost is the recommended final forecasting model because it provided the strongest overall predictive accuracy and captured nonlinear relationships among historical demand, temporal behavior, and weather conditions.

From an internship perspective, the project demonstrates an end-to-end data-science workflow: data acquisition and cleaning, exploratory analysis, statistical modeling, machine-learning development, feature engineering, model evaluation, explainability, output generation, GitHub project organization, and dashboard development. It provides a solid foundation for a more scalable operational forecasting system using longer historical records, richer exogenous variables, systematic validation, and live deployment.

Appendix A - Project Notebook Structure

Notebook	Scope
<i>01_energy_data_exploration.ipynb</i>	<i>CAISO demand exploration, decomposition, stationarity testing, SARIMA, Prophet, Hybrid forecasting, and Phase 1 evaluation.</i>
<i>02_noaa_weather_integration.ipynb</i>	<i>Weather integration, feature engineering, weather-only Random Forest, Enhanced Random Forest, XGBoost, feature importance, and Phase 2 evaluation.</i>

Appendix B - Key Output Files

- *forecast_results.csv* - Phase 1 actual demand and statistical-model forecasts.
- *model_comparison.csv* - consolidated model performance metrics.
- *xgboost_predictions.csv* - XGBoost prediction output.
- *xgboost_feature_importance.csv* - ranked XGBoost feature importance.
- *dashboard/app.py* - Streamlit dashboard application.